



FINAL PROJECT

DATABASE

FOR A RIDE SHARING APPLICATION

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PROJECT OVERVIEW

GOAL:
BUILD A DATABASE FOR AN
APP LIKE UBER.



MAIN ELEMENTS



Persons



Cars



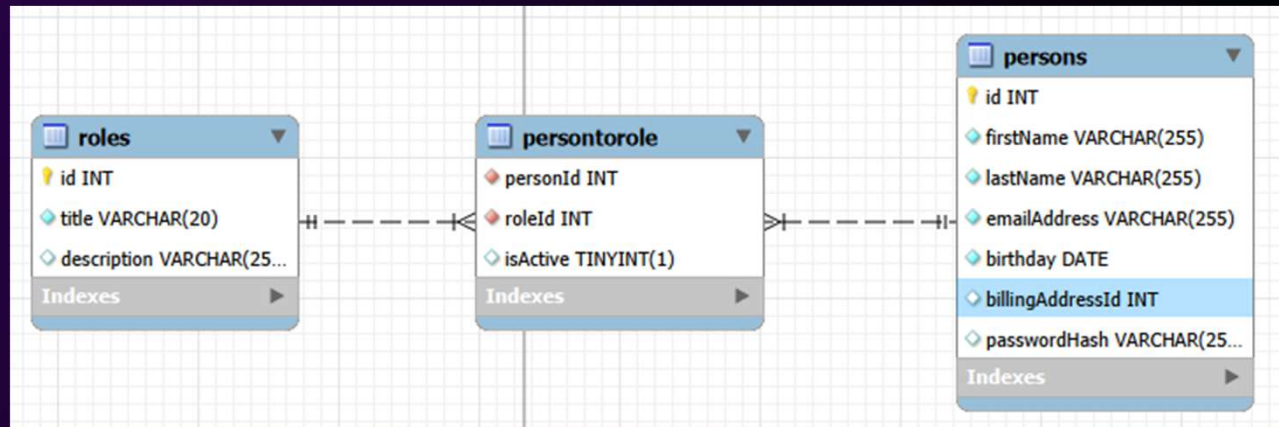
Addresses



Rides



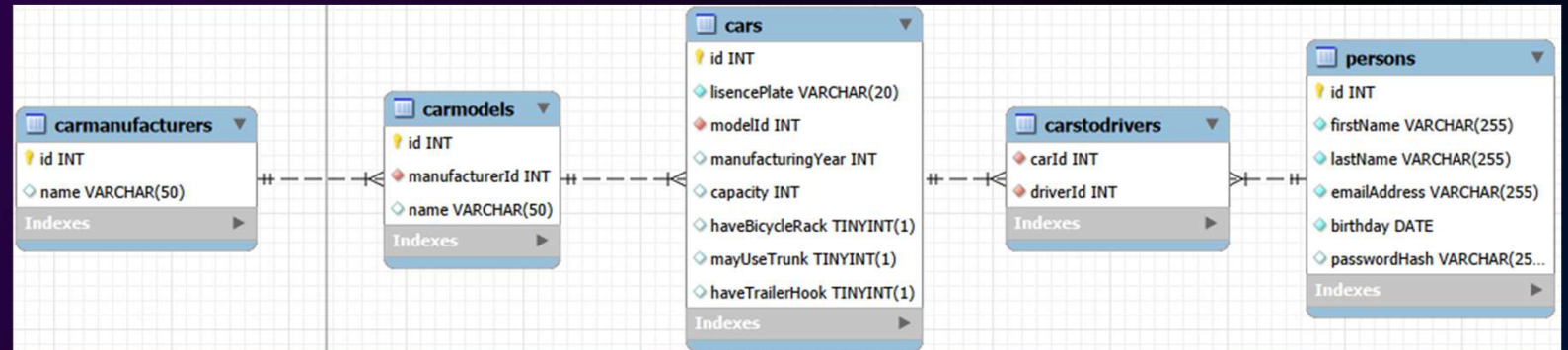
Persons



- Roles: Admin, Passenger, Driver
- Many to many between persons and roles , via person_To_Role table
- Persons table includes basic persons data
 - Name
 - Email address (used as username too)
 - Password (hashed)
 - ...



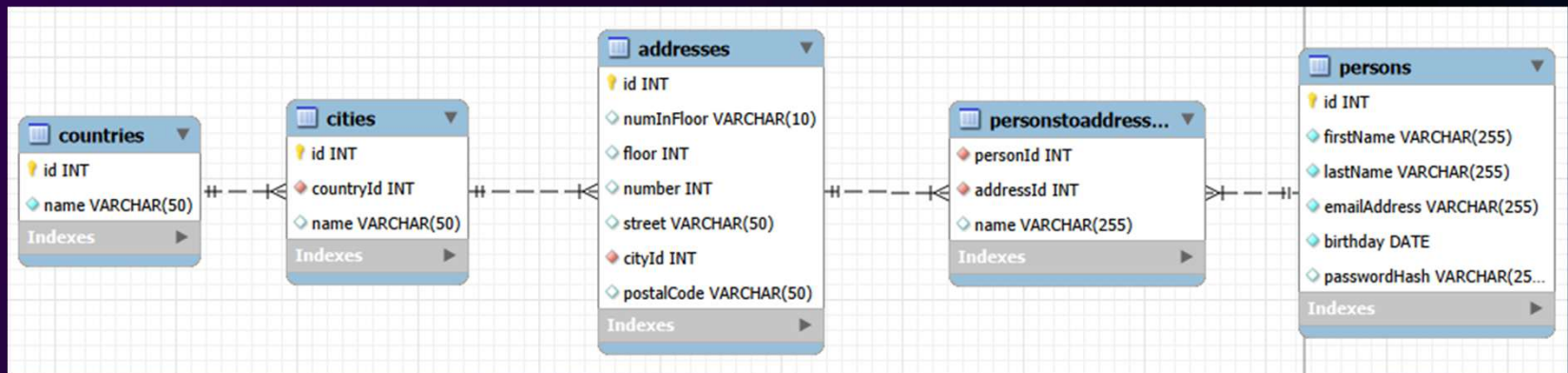
Cars



- One to many relationship between car manufacturers and car models
- One to many relationship between car models and car
- Many to many relationship between cars and persons, via cars_to_drivers table.
 - A person can be related to many cars
 - For example, a young child that can borrow car from both parents
 - The same car can be used by several drivers
 - For example, a couple that both work for uber but they have only one car.



Addresses



- One to many relationship between countries and cities
- One to many relationship between cities and addresses
- One to many relationship between cities and addresses.
- Many to many relationship between address and persons

Future development:

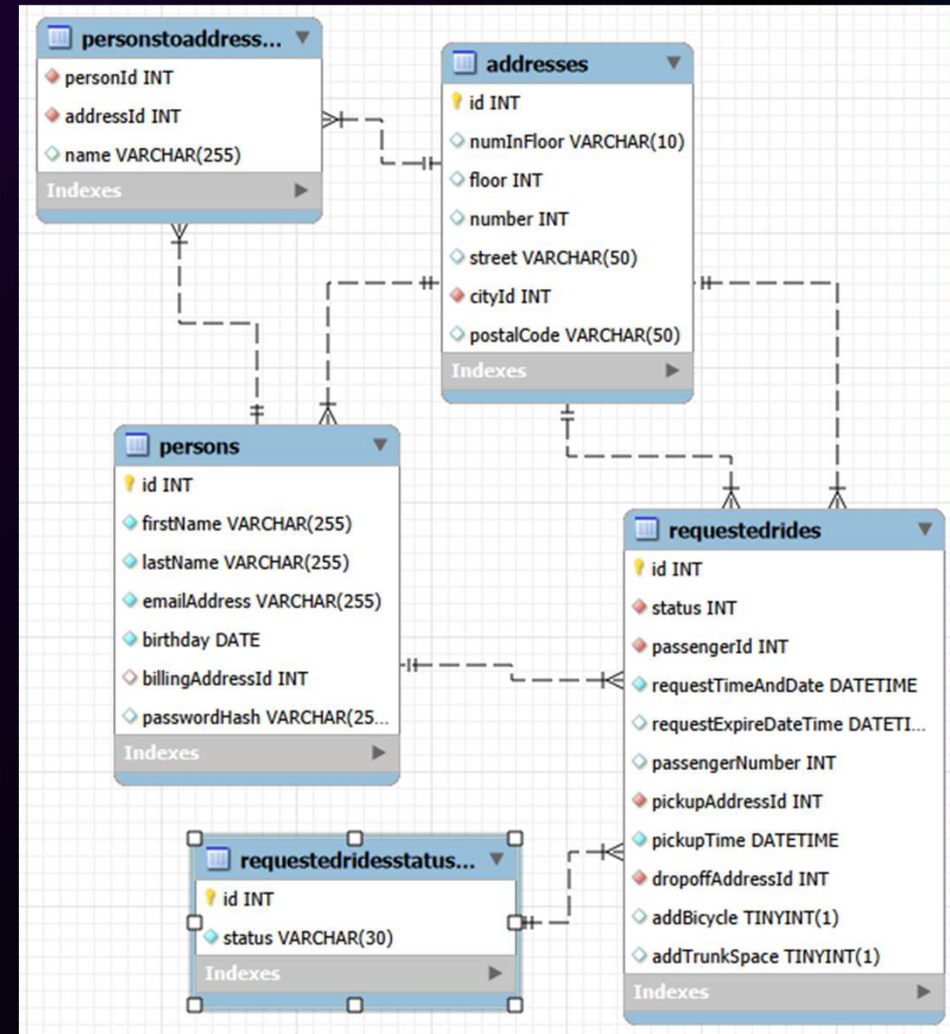
Make a table for streets with a foreign key to cities.

Make a table for building number and use street ID as a foreign key



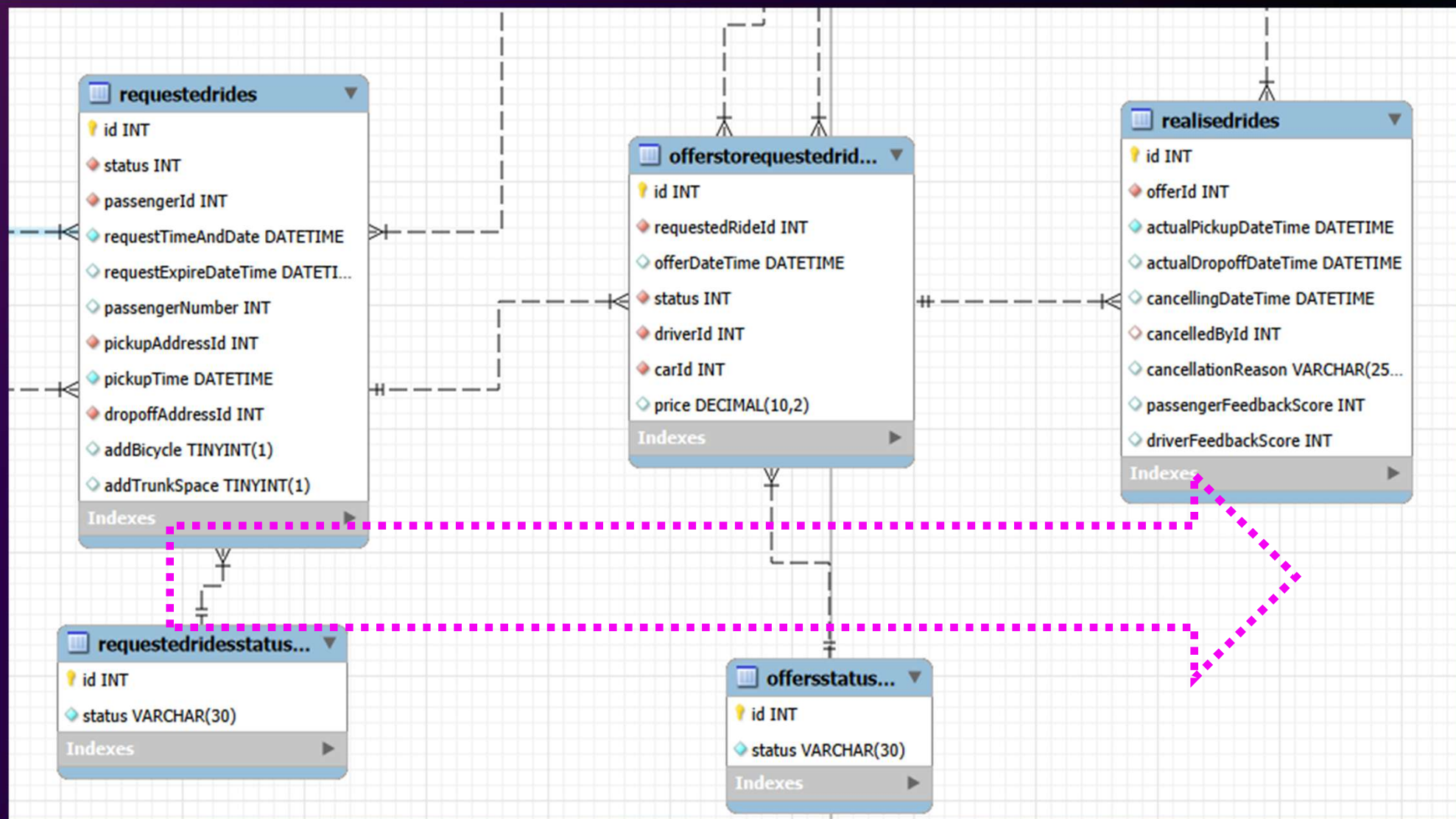
Rides

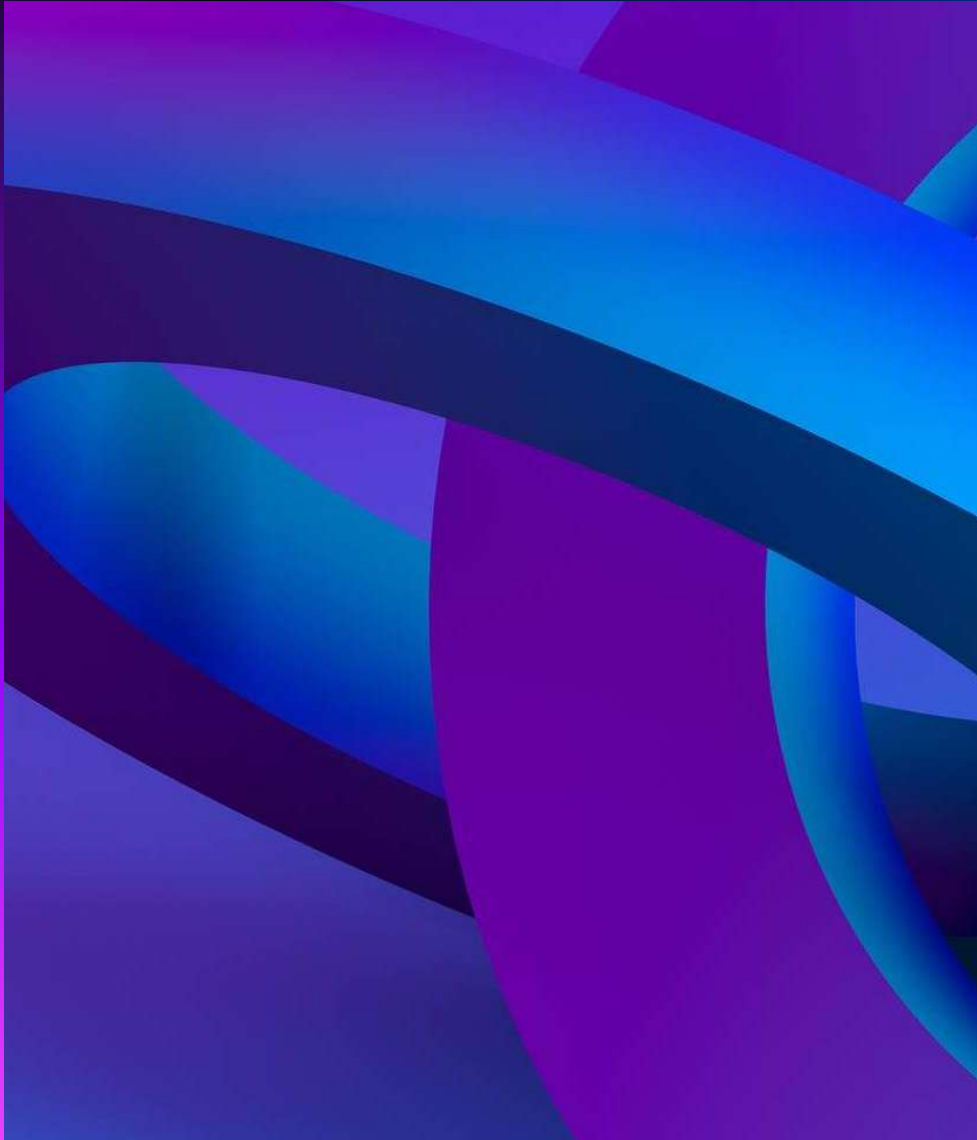
- Passengers registers a ride request
- Each ride refers to passenger ID, pickup address ID and drop off address ID
- Each ride has a status:
 - Pending
 - Expired
 - Accepted
 - Cancelled
- Each ride request has an expire time-date.



LIFECYCLE OF A RIDE

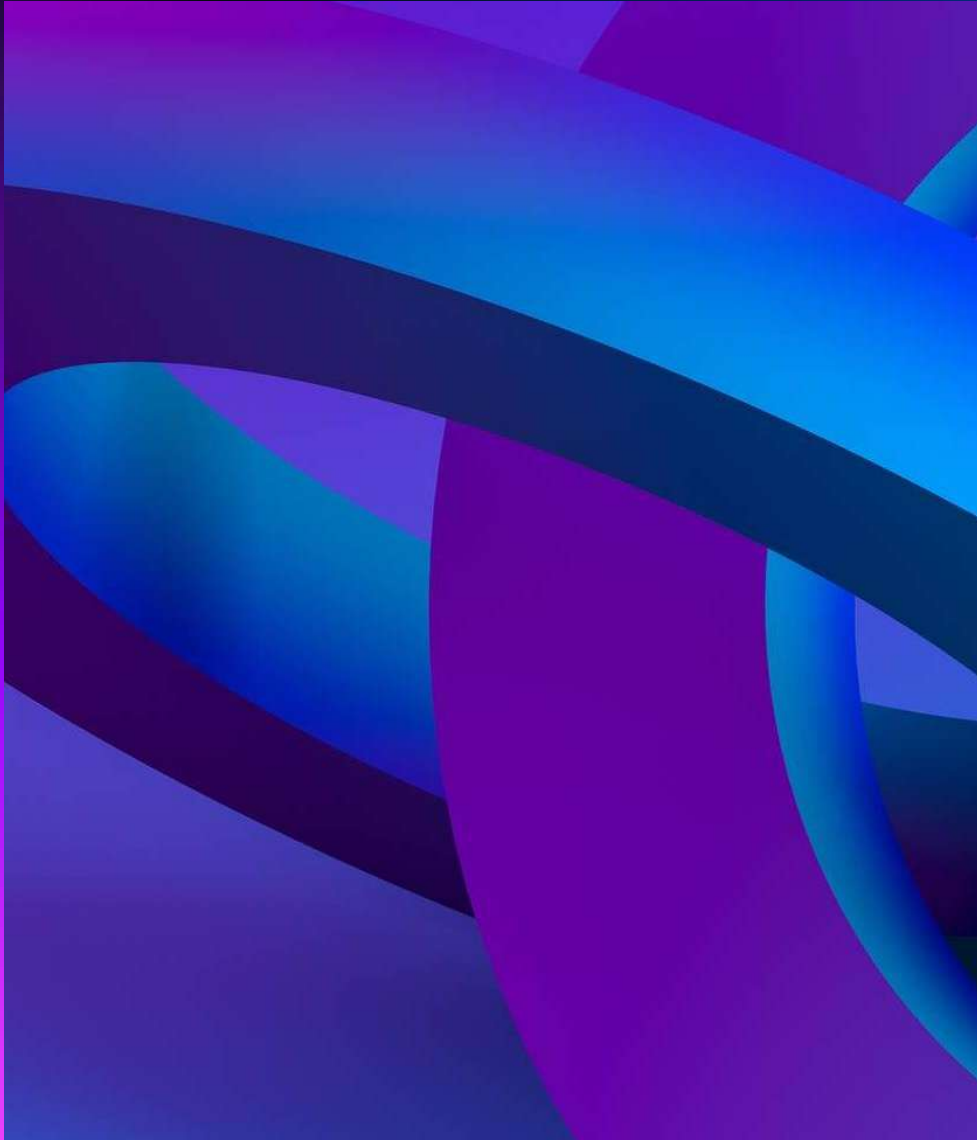






ACHIEVEMENTS

- An atomic, consistent, Isolated, and durable database
- Built based on NF3
- Automated trigger for changing status of expired requests
- A very good learning experience by having hand on the job



POSSIBLE IMPROVEMENTS

- Make a separate table for street names
- Make a separate table for the house numbers for each street
- Make tables for payments and credits
- Make a tables for messages between users
- Make a table for logs